Salvador Robles

salvadorroblesherrera@gmail.com | salvadorrobles10 | salvadorrh | 915 282 6436

Education

The University of Texas at Austin — Expected: Dec 2026
Master of Science in Computer Science ***(GPA: 3.83 / 4.0)*** — *Austin, Texas*

- **Courses:** NLP, Visual Recognition, Generative Models, Distributed Systems, Parallel Algorithms, Convex Optimization

The University of Texas at El Paso — Graduated: May 2024
Bachelor of Science in Computer Science and Mathematics ***(GPA: 3.94 / 4.0)*** — *El Paso, Texas*

- **Awards:** Department Rank 1 (Mathematics) **Academic Award Scholarship (top 0.1%)**
- **Courses:** Machine Learning, Deep Learning, Computer Vision, Linear & Abstract Algebra, Statistical Inference

Experience

Pinterest | *PyTorch, PySpark, Apache Airflow, AWS S3, Learning-to-Rank, A/B Testing* — *San Francisco, California*
Master's Machine Learning Engineer Intern — *May 2026 - Aug 2026*

- Developed and shipped personalized notification title ranking models, increasing **270K+ WAU** and **97K+ MAU**.
- Boosted engagement by **0.8%** for Push and **1.0%** for Email notifs by creating separate models for each channel.
- Evaluated **200+ neural networks** across **240M+** examples, finding models that improved production performance.
- Built **Spark + Airflow** pipelines for large-scale training and evaluation, serving online models to **600M+** users.

The University of Texas at Austin | *Python, PyTorch, NumPy, Pandas, Multi-Agent systems* — *Austin, Texas*
Teaching Assistant - Courses: NLP, AI Agents, and Data Analytics — *Aug 2024 - May 2026*

- Designed and graded **30+** Python assignments for **60+ students**, covering ML, transformers, LLMs, and agents.
- Tutored a total of **100+** students in bi-weekly office hours, earning **100% positive feedback** for teaching clarity.

Uber | *Python, Apache Spark, Scala, Pandas, Data Security* — *Seattle, Washington*
Machine Learning Engineer Intern — *May 2022 - Aug 2022*

- Integrated ML models to classify and flag sensitive user data (e.g., identity + location) in Uber's DataK9 platform.
- Boosted accuracy from **87% to 93%** on exabyte-scale data, enabling categorization of over **400K+** datasets at Uber.
- Reduced latency by **12%** by developing a Scala-based **Apache Spark** pipeline to collect training data for inference.
- Designed and evaluated **10+** feature engineering approaches, identifying key predictors for **6%** accuracy gain.

Google | *C++, Python, Vertex AI, Recommendation Systems* — *Waterloo, Ontario (Remote)*
Software Engineering Intern — *May 2021 - Aug 2021*

- Contributed to Cloud AI's large-scale ML workload migration from Borg to **Vertex AI**, supporting **1k+ pipelines**.
- Engineered a multi-threaded **C++** handler, tuning concurrency across **5–30 threads** and improving throughput by **12%**.
- Developed and tested support for scalable vector matching relying on a container's local volume in **Python**.

Google | *JavaScript, Java, Google Maps API, HTML/CSS, UI/UX, REST APIs* — *Mountain View, California (Remote)*
Student Training in Engineering Program Intern (STEP Intern) — *May 2020 - Aug 2020*

- Built a full-stack route planner using Google Directions API and a grid-based datastore modeling **500 crime zones**.
- Built a JavaScript obstacle-avoidance algorithm that evaluated **20+ routes** and returned the **top 3 safest routes**.

Projects

Disaster Response Message Classification | *Fine-Tuning BERT, RAG, FastAPI* — *Dec 2025*

- Deployed a **FastAPI web application** for multi-label disaster message classification using a fine-tuned BERT model.
- Integrated a **FAISS-based RAG pipeline**, indexing **26k+** messages and achieving **Recall@5 = 0.85** for semantic retrieval.

Benchmarking Spatial Reasoning in Vision-Language Models | *PyTorch, Transformers, OpenCV* — *May 2025*

- Benchmarked spatial reasoning in Vision-Language Models, focusing on left–right consistency, and size perception.
- Evaluated SOTA models (**LLaVA-7B/13B, GPT-4o, Claude-3.5**) on **5k+** COCO images and **720** synthetic scenes.

Publications and Activities

- Predicting Fairness of ML Software Configurations — PROMISE'24 **(Presented in Top Conference)**
- When is Deep Learning Better and When is Shallow Learning Better — IJPEDS'22 (**Best Student Paper**)
- **Officer, The Coding Interview Club**: Coached **50+** students to land internships via mock interviews and weekly sessions.
- **Clubs:** UT Competitive Programming, Hispanics in Computer Science (HACS), Texas ACM, ML and Data Science Club.
- **5-time National Medalist** (Silver/Bronze, **Top 3, 0.001%**) in the Mexican Mathematics Olympiad.

Languages and Technologies

Languages: Python, C/C++, Java, Scala, JavaScript, R, HTML/CSS, Haskell
Systems & Backend: AWS, GCP, Docker, Kubernetes, Linux, Git, Bash, CI/CD, Airflow, Flask, FastAPI, Node.js
Data, ML & AI Tooling: SQL, Apache Spark, PySpark, MongoDB, PyTorch, Pandas, NumPy, Claude Code, Cursor.